LCAForge

In-depth summary of the project

Materials intelligence and life cycle assessment for Tunisian exporters · Prepared 24 October 2026

Reading guide: figures in square brackets point to the numbered sources in Appendix A. "Modeled" means an internal estimate, not a measured result. Project-status statements come from the founder's own notes and should be re-checked before any external use.

1. Executive summary

LCAForge is a software platform that turns a manufacturer's own data (energy, materials, process steps, suppliers) into carbon footprints, life cycle assessments (LCA) and reports. It is designed for Tunisian industrial companies that export, above all to Europe, where the Carbon Border Adjustment Mechanism (CBAM) has been in its definitive, paid phase since 1 January 2026 [1][2].

The core idea is simple. When an exporter cannot document its real emissions, the EU applies default values with a markup of 10% in 2026, rising to 30% from 2028 [3]. Measured, verified data replaces those defaults and lowers the bill that the European importer pays, and then passes back to the supplier [3]. LCAForge aims to make that measurement fast, repeatable and affordable, and to reuse the same data for eco-design, product LCA, costing, R&D decisions and supplier risk.

What makes it distinctive: (1) one platform with six modules instead of a single-purpose LCA tool; (2) a Copilot agent that drafts a plan from a plain-language request, waits for the user to edit and validate it, then executes it visibly inside the application and exports a report; (3) a focus on Tunisian exporters, with local support and prices in dinars.

Where it stands: the project notes describe a working platform with a shipped training module, a filed patent and positive feedback from a CNRS lab head and SOPAL engineers; the new web dashboard, the Copilot's visible execution and the integration of free LCA databases were still being built on 24 October 2026 (see section 8).

2. The problem and the market context

2.1 Carbon is becoming an export cost

CBAM puts a carbon price on selected imports into the EU. It currently covers iron and steel, cement, aluminium, fertilisers, electricity and hydrogen [4]. The importer buys CBAM certificates priced from EU ETS auctions: €75.36 for Q1 2026 [1], and €82.32 for Q3 2026 [2]. Certificates for 2026 imports are bought from February 2027 [1]. EU buyers who pay more because of default values are expected to push that cost back to their suppliers [3].

Other jurisdictions are following at different speeds [2]:

Jurisdiction	Status (October 2026)	Sectors	Source
European Union	In force since 1 Jan 2026; certificate price published quarterly, weekly from 2027	Steel, cement, aluminium, fertilisers, electricity, hydrogen	[1][2][4]
United Kingdom	Law (Finance Act 2026); starts 1 Jan 2027; illustrative rates GBP 1.45–9.79 per tCO_2e, no binding rate yet	Aluminium, cement, fertilisers, hydrogen, iron and steel	[2]

Jurisdiction	Status (October 2026)	Sectors	Source
Norway (and Iceland)	EEA decision extends the EU mechanism from 1 Jan 2027; not yet in force on 13 Oct 2026	As EU	[2]
Australia	Border adjustment recommended (Feb 2026 review), under consideration; not legislated	Cement and clinker first	[2]
New Zealand	Minister signalled a mechanism aligned with Australia; nothing proposed	n/a	[2]
Canada, Japan, South Korea	No border measure proposed (domestic carbon pricing only)	n/a	[2]

The EU is also negotiating a wider list of downstream products (estimates range from about 180 to 457 products); a first trilogue on 20 October 2026 ended without agreement, and the target application date is January 2028 [2]. This is still a proposal.

2.2 Why real data lowers the bill

Default values reflect the emissions of the worst-performing facilities, plus the markup [3]. Using actual emissions removes the markup and requires an annual audit by an accredited verifier [3]. The vendor blog below gives two illustrative calculations, using €75 per tonne of CO_2:

Example	Default (with markup)	Actual	Bill 2026 (default)	Bill 2026 (actual)	Saving 2026
Steel slab, 10,000 t	3.483 tCO_2/t	1.800 tCO_2/t	€65,319	€33,750	€31,569 (48%)
Portland cement, 50,000 t	1.742 tCO_2/t	0.880 tCO_2/t	€163,350	€82,500	€80,850

Source [3], a software vendor's marketing blog, so treat it as illustrative. The bills in these examples are only about 2.5% of the gross carbon cost (my calculation from the blog's own figures), because the CBAM obligation is phased in gradually. The same blog projects that the saving on 10,000 t of steel grows to about €842,848 per year by 2030 as the phase-in factor rises. In 2026 the saving per tonne is therefore only a few euros, not tens of euros.

2.3 The Tunisian context

- **National target:** reduce the carbon intensity of the economy by 45% by 2030 compared with 2010 [4].
- **ANME:** in August 2026 the energy agency published a carbon-accounting specification covering company GHG inventories, product carbon footprints and CBAM support [4].
- **Fonds de transition énergétique (FTE):** support of up to 70% of the cost of the accompaniment, capped at 70,000 TND [4].
- **EU programmes:** TCP (grants, risk-sharing, advice) and GEFF Tunisia (green credit lines through local banks) [5].

Whether LCAForge itself qualifies for any of these schemes has not been confirmed and should be checked with ANME before it is promised to a client.

3. The product

3.1 Six modules, one data foundation

Module	What it does	Example target companies (from the founder's client directory)
M1 Carbon and processes	Carbon footprint by process step; CBAM-ready data	SKG Group, SAT Sfax, Hammami Transport
M2 Eco-design	Compare materials before production	MECAFORM, Baplast, MATEM Group, EKM Bois, SIMP
M3 Product LCA	Product-level LCA for customers and tenders	EPImed, Laboratoires Nihel
M4 Cost and materials	Reduce material cost, scrap and energy	MIM, AMI, ETAM, SNCM, Sfax Fonderie CNC
M5 R&D decisions	Choose the best option faster	SACOGES, GIMO
M6 Suppliers and risk	Spot at-risk suppliers before a disruption	GMP Construction, Sfax Cables, Alpha Polycom

3.2 Data layer

- **Integrated and verified (per the 13 October technical summary):** EF 3.1 and BONSAI.
- **In progress:** USLCI, and the wider integration of free LCI, LCIA and materials-AI databases listed in the founder's reference workbook (planned as "Wave 2D").
- **Blocked or not licensed:** BAFU, ecoinvent and TianGong connectors. ecoinvent is not included, which matters because many LCA practitioners expect it.

3.3 The Copilot agent

The Copilot is a chat panel for end users. The user describes a need in plain language; the agent drafts a task plan; the user edits and must confirm the plan; only then does it run. Execution is meant to be visible: pages change, inputs are filled and calculations run in the application, and the results are exported as a PDF or another format. AI is deliberately limited to two jobs, drafting the plan from the user's prompt and writing reports (with retrieval that places the platform's own results directly into the report); the calculations themselves are not delegated to a language model. The notes record a local model setup for first tests rather than a hosted service.

3.4 Platform and engineering

- **Training module:** a five-tab workflow (Data, Prepare, Train, Evaluate, Models) accepting CSV, XLS or manual entry, with per-tenant model training, resource guardrails (RAM, VRAM, CPU, dataset size, time) and tenant-isolated model storage. The notes record 321 tests passing at the time.
- **Interface:** the Streamlit application remains the legacy UI until the Next.js dashboard reaches parity and passes end-to-end and Electron checks; Electron wraps the same Next.js app. A GUI redesign for accessibility is under way.
- **Prediction pipeline and IP:** the AI materials-property prediction system is the subject of a patent filing with INNORPI (submitted, pending examination as of September 2026). Its architecture is intentionally not described in this summary.

4. Value: time and money

4.1 Time

The pitch material states that an LCA falls from about 493 minutes (more than 8 hours) to about 9 minutes with an automated workflow, roughly a 98% reduction. The figure comes from the founder's source document, attributed to a 2025 Elsevier BIM-LCA study about buildings. I found the likely paper [9] but could not confirm the two numbers in the text available, so check them in the article or soften the claim to "minutes instead of hours".

4.2 Money: real cases (vendor-reported)

Company	Reported result	Tool	Source
Interface (carpet tiles)	Energy use −40% and yarn per unit −12% since 1996; total annual cost savings of €7.6m, as part of a programme supported by LCA	GaBi (thinkstep/Sphera)	[6]
Carrier (HVAC)	One plastic part change cut annual costs by $60,000 and embodied carbon by 10 t	aPriori	[7]
ERSOS (heat pumps)	$671 per unit cost reduction and 19.32% CO_2 reduction from a few material and supplier substitutions	Makersite	[8]

These are published by the software vendors themselves and describe different companies and sectors. They show what LCA tooling can achieve; they are not guarantees for a Tunisian SME.

4.3 Money: modeled ROI for LCAForge

Client profile	Annual subscription	Modeled net annual gain	ROI	Payback
Small lab	9,600 TND	+11,700 TND	122%	~5 months
Industrial SME	48,000 TND	+126,500 TND	264%	~3 months
Intensive exporter	60,000 TND	+219,000 TND	365%	~3 months

Internal model based on literature and the founder's assumptions; not measured on a real client. The first pilot should be used to replace it with real numbers, especially for modules M3, M5 and M6, where no verified saving exists yet.

5. Competitive landscape

Competitors fall into three groups: LCA specialists, integrated product-sustainability platforms, and adjacent tools for materials AI, supplier risk and simulation. Public price data exists for only some of them.

Tool	Scope	Public yearly price found	Source
SimaPro	Professional LCA software	Expert licence about €6,100 (reseller price)	[10]
One Click LCA	Product LCA and EPD; building LCA	From about €4,800 for product LCA (€400 per seat per month, billed annually)	[11]
Ecochain Mobius	Product LCA	Starter from €95 per month; Professional €340 per month (about €1,140 to €4,080 per year)	[12]

Tool	Scope	Public yearly price found	Source
Makersite	Materials, LCA, cost and supplier risk; enterprise platform	Quote-based	[13]
Sphera GaBi	Professional LCA software	Quote-based (not checked)	—
openLCA	Open-source LCA	Free software; databases and support cost extra (from my own knowledge, not checked)	—
EcoVadis	Supplier sustainability ratings	$500 to $11,000+ per year (third-party estimate)	[14]
Citrine, Materials Zone, Matlantis, Albert, Schrödinger, Ansys, BIOVIA, COMSOL, Simcenter, Resilinc, Interos, Prewave	Materials AI, simulation, supplier risk	Quote-based or academic-only pricing (not verified)	—

Positioning. The closest overall competitor is Makersite, which also combines materials data, LCA, cost and supplier risk, but it is an enterprise platform. The founder's documents state that no integrated software is designed for Tunisian manufacturers; LCAForge's advantage is breadth, local support, dinar pricing and a CBAM focus. It does not currently have ecoinvent, and it should not claim to be cheaper in every tier: the industrial-SME subscription (about 42,000 to 60,000 TND per year) is likely above SimaPro Expert or One Click LCA once converted at the current exchange rate.

6. Business model and go-to-market

6.1 Pricing (founder's current grid)

Offer	Price	Notes
Pilot, 3 months	1,500 TND	Versus 10,500 to 15,000 TND at normal rates; on the client's real data
Laboratory plan	800 to 1,200 TND per month	10% discount for annual payment
Industrial SME plan	3,500 to 5,000 TND per month	10% discount for annual payment

6.2 Target clients

The initial target is the Sfax industrial zones (the notes cite more than 80 manufacturers across six sectors in the Oued Chaabouni zone). The directory is being widened from the original 20 firms to all Sfax industrial areas (Oued Chaabouni, Route de Gabès, Poudrière 1 and 2 and others), with free and paid database needs recorded per client. The founder's internal fit scores for the top candidates:

Company	Fit score (/10)	Why it fits (module)
SKG Group	10	M1 carbon and processes; a discovery questionnaire (25 questions, French and English) is ready
MECAFORM	9.5	M2 eco-design; sheet-metal and automotive components
EPImed	9.5	M3 product LCA

Company	Fit score (/10)	Why it fits (module)
Baplast	9	M2 eco-design
SAT Sfax	9	M1 carbon and processes

Scores are the founder's own estimates, not measured. Whether a given client's products fall inside the CBAM scope depends on its customs codes and should be checked per client.

6.3 Sales material and the booth

Pitch decks exist for three audiences (industrial companies, banks, SICARs and investors), with French and English speaker notes. For the booth the plan is a live demonstration of the Copilot rather than printed flyers, backed by a short HTML presentation that runs on a loop and cites its sources.

7. Company, funding and validation

- **Founder-led:** the platform is built by its founder, who is also co-founder and CEO of MINI AI Tech and holds a master's degree in condensed matter physics. Planned hires: ML/AI engineers and full-stack developers first; later business development, finance and administration, industrial support and LCA/regulatory expertise.
- **Legal:** LCAForge is being formed as a SARL (documents with an accountant). The business plan's top recommendation is to apply for Startup Act status.
- **Business plan:** a nine-stage guided plan (opportunity, market, team, commercial strategy, operations, investment, projections, Monte Carlo risk simulation, assembly) with a three-year horizon is complete.
- **Funding:** a bridge ask of 17,000 to 47,000 TND appears in the bank and investor decks. A separate IPTIC request of 10,000 TND plus 5,000 TND from the founder (15,000 TND in total) is planned to buy a workstation PC (quote of 8,200 TND including tax) for MINI AI Tech.
- **Validation so far:** positive, concrete feedback on the need from a CNRS lab head and from engineers at SOPAL. No paying client is recorded in the notes.

8. Status and roadmap

Area	Status according to the founder's notes
Shipped	Training module overhaul (321 tests passing); patent filed with INNORPI; business plan complete; 25-question SKG questionnaire; three pitch decks and speaker notes
In progress (24 Oct 2026)	Next.js dashboard with the Copilot chat panel (visible execution after validation); integration of free LCI/LCIA/materials-AI databases; local model profile for report writing and plan drafting; master prompt v2.2 prepared but not yet run in the coding agent
Roadmap position	The 13 October technical summary placed the platform at wave 3.5 of 8; it may be out of date
Open dependencies	ecoinvent licence; BAFU and TianGong connectors; real-client pilot data; training results (ML training on Kaggle is slow, results to be added later)

For the booth, the presentation says "functional, ready to use". If the demonstration is complete end to end, say so; if any part of the Copilot's visible execution is still unstable, prepare a recorded fallback and avoid the word "finished".

9. Risks and limitations

Risk	Mitigation or note
No ecoinvent licence	Use EF 3.1, BONSAI and other free databases; state the limitation openly; add paid databases per client need
Credibility of savings claims	Label vendor case studies and the modeled ROI clearly; replace them with pilot results
CBAM scope and verification	Check each client's customs codes; data must still be verified by an accredited verifier, so LCAForge prepares data rather than replacing the verifier
Eligibility for Tunisian aid	Not confirmed; ask ANME before promising co-funding
Pricing pressure	International tools cost €1,100 to €6,100 per year; compete on breadth, local support and CBAM focus; offer a low-cost pilot
Small team	Hire engineers first; keep the scope of each release narrow; use the pilot to prioritise
Patent pending	Keep model architecture out of public material until examination progresses
Demo risk at the booth	Test the full Copilot path beforehand; keep a recorded fallback and a looping slide deck

Appendix A. Sources

[1] CBAM certificate price Q1 2026 (€75.36), purchases from Feb 2027 — PwC Austria, 9 Apr 2026. https://www.pwc.at/de/insights/austrian-tax-news/cbam-certificates-eu-commission-price.html

[2] Q3 2026 price (€82.32); UK, Norway, Australia, New Zealand, Canada, Japan, South Korea; EU downstream extension — CBAM Guide, updated 23 Oct 2026. https://cbamguide.com/trade/global-cbam/

[3] Default values vs actual emissions, 10/20/30% markup, steel and cement examples — The Sustainability Cloud (software vendor blog), 3 Sep 2026. https://www.thesustainabilitycloud.com/blog/eu-cbam-default-values/

[4] FTE (70% up to 70,000 TND), ANME carbon-accounting specification (Aug 2026), CBAM scope, national target — Tunisie Numérique, 26 Aug 2026. https://www.tunisienumerique.com/decarbonation-entreprises-tunisiennes-financement-transition-energetique/

[5] GEFF Tunisia credit lines and TCP programme — WMC with TAP, 1 Dec 2025. https://www.webmanagercenter.com/2025/12/01/556948/tunisie-ue-programmes-europeens-et-financements-verts-pour-preparer-les-entreprises-au-mecanisme-carbone/

[6] Interface case study (energy −40%, yarn −12%, €7.6m per year) — thinkstep/Sphera via csr.dk, 2018. https://www.csr.dk/opening-new-markets-and-achieving-corporate-objectives-lca

[7] Carrier case study ($60,000 per year, 10 t embodied carbon) — aPriori via CaseStudies.com. https://www.casestudies.com/company/apriori/case-study/carrier-saves-6-months-on-lca-process-with-apriori

[8] ERSOS case study ($671 per unit, 19.32% CO_2) — Makersite with Durham University. https://makersite.io/?p=3387

[9] BIM-LCA systematic review, Building and Environment 282 (2025) 113248, Elsevier (the 493 → 9 minute figure is not confirmed in the text I could access). https://repositorio.iscte-iul.pt/bitstream/10071/34667/1/article_111613.pdf

[10] SimaPro Expert licence price — SimaPro reseller (Sweden). https://simapro.se/en/?p=4527

[11] One Click LCA product LCA / EPD pricing. https://oneclicklca.com/pricing/product-lca-and-epd-pricing

[12] Ecochain Mobius FAQ and pricing page. https://helpcenter.ecochain.com/en/articles/9059735-mobius-faq

[13] Makersite profile (enterprise platform, B2B sales) — KI-Syndikat. https://www.ki-syndikat.de/tools/makersite/

[14] EcoVadis pricing estimate ($500 to $11,000+ per year) — Dcycle. https://dcycle.io/blog/ecovadis-pricing-plans-medals

Items that come from the founder's own documents (pricing grid, modeled ROI, fit scores, funding figures, project status) have no external source and are internal data.

Appendix B. Corrections made while preparing this material

- An earlier draft said real data saves about €65 per tonne of cement. That figure ignored the CBAM phase-in; the 2026 saving in the vendor's own example is about €1.6 per tonne (€80,850 on 50,000 t). The pitch decks that still show €65 per tonne should be replaced by the latest version.
- An earlier draft said Canada, Australia and Taiwan were preparing their own border mechanisms. The current source says Canada has proposed none and Australia is reviewing a recommendation; Taiwan is not mentioned there. The corrected list is in section 2.1.
- The certificate price was updated from the Q1 figure (€75.36) to the latest published quarter (Q3 2026: €82.32).